

Empirical Evidence Regarding Relations Among a Model of Epistemic and Ontological Cognition, Academic Performance, and Educational Level

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Models of personal epistemology have not been sufficiently integrated despite conceptual similarities. We attempted to model both dimensional and positional aspects of personal epistemology, as well as examine the domain specificity of these phenomena. The conceptual framework for this study was a new model of epistemic and ontological cognitive development that addresses numerous concerns in the literature. A model-based quantitative instrument was created and administered to 740 students ranging from middle-school through graduate school. Results of confirmatory factor and factor-mixture model analyses provided evidence for the construct validity and reliability of scores from the instrument as well as support for many aspects of the underlying conceptual model. Implications for further model development and research are discussed.

Keywords: personal epistemology, epistemological beliefs, epistemic cognition, philosophy, factor mixture modeling

Researchers have provided evidence that individuals' personal epistemology, their beliefs about knowledge and knowing (Hofer & Pintrich, 1997), can influence their learning, problem solving, intensity of mental effort, and academic success (Kuhn, 2005; Schommer-Aikins & Easter, 2006). Models of personal epistemology have generally come in two forms: (a) a "developmental" stage, or position, model with qualitatively different subtypes (King & Kitchener, 2004; Kuhn, Cheney, & Weinstock, 2000; Perry, 1970) or (b) a "system of beliefs" dimensional model that characterizes individuals in terms of quantitative differences along multiple belief continua (Hofer, 2004a; Hofer & Pintrich, 1997; Schommer-Aikins, 2004). Yet, there is considerable overlap be-

tween these two types of models (Bendixen & Rule, 2004; Hofer, 2004b; Hofer & Pintrich, 1997; Muis, Bendixen, & Haerle, 2006). Hofer (2001) suggested that personal epistemology research may be moving toward "an integration of ideas from multiple models: an identifiable set of dimensions of beliefs, organized as theories, progressing in reasonably predictable directions, activated in context, operating as epistemic cognition" (p. 377).

A key aspect of Hofer's (2001) call for integration is the idea that both developmental positions and systems of beliefs dimensions may be necessary to accurately capture students' personal epistemology. To some degree, certain developmental models do address both positions and dimensions, with analysis of qualitative data regarding the latter used to indicate the former (Baxter Magolda, 2004; Perry, 1970). As an example, in King and Kitchener's (1994, 2004) research, qualitative interviews were conducted to determine participants' beliefs along numerous dimensions, with the pattern of beliefs used to characterize participants as being in one of seven stages. However, the focus was on positions and not on how dimensional beliefs develop over time. Researchers who investigate models using quantitative methods have focused on dimensional beliefs but have not examined whether those beliefs, as a group, progress "in reasonably predictable directions" (Hofer, 2001, p. 337) indicative of overarching stages or positions. This may be because quantitative investigations of personal epistemology have failed to produce scores that meet standard criteria for reliability and validity (Buehl, 2008; Clarebout, Elen, Luyten, & Bamps, 2001; DeBacker, Crowson, Beesley, Thoma, & Hestevold, 2008; Wood & Kardash, 2002). We know of no conceptual model that focuses on capturing relations between positions and dimensions simultaneously while also examining other covariates of interest such as grades or educational level.

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Our research had four interrelated goals. First, we wished to answer Hofer's (2001) call by developing a conceptual model of personal epistemology that puts forward specific predictions regarding how quantitative differences in dimensional beliefs can be used to characterize individuals as being within qualitatively different positions. Second, we sought to evaluate whether beliefs about the nature of knowledge and knowing can vary across domains (e.g., domain generality vs. some degree of specificity). Third, as King and Kitchener (2004) suggested, we hoped to "bootstrap" (p. 11) between theory development and instrument refinement, iterating toward an integrated and testable conceptual model. We hoped this bootstrapping would lead to a fourth goal, a survey instrument that produces scores that are reliable and valid indicators of personal epistemology.

This article describes the development and testing of a quantitative measure of a model of epistemic and ontological cognition (EOC; Greene, Azevedo, & Torney-Purta, 2008). This model specifies relations between developmental positions and systems of belief dimensional aspects of EOC and is informed by philosophical epistemology. By surveying participants ranging from middle school to graduate students, we hoped to provide empirical support for the structure and relations posited in the underlying conceptual model, including relations among positions, academic performance, and educational level. We also sought to provide "proof of concept" that both qualitative and quantitative differences in EOC can be modeled simultaneously and that these models can provide a better fit to the data than ones focusing solely on dimensional beliefs.

Developmental and Systems of Beliefs Models of Personal Epistemology

Perry's (1970) original conceptualization of personal epistemology included qualitatively different positions characterized by students' views regarding knowledge, knowing, and the role of authority, among other factors. This conceptualization laid the groundwork for numerous subsequent developmental models of personal epistemology (e.g., Baxter Magolda, 2004; King & Kitchener, 1994, 2004; Kuhn et al., 2000). These models are reviewed extensively elsewhere (e.g., Hofer & Pintrich, 1997, 2002). Although the language and descriptions differ slightly, generally each model outlines a similar set of stages, positions, or levels, summarized here with Kuhn et al.'s (2000) terminology (see Table 1). In these developmental models, individuals begin with an objective view of the world where knowledge is certain and often derived from authority figures, reducing thinking to the act of determining the "correct" answer from among the available options. With development, individuals are posited to move into a

multiplist level, where knowledge is seen as entirely subjective, making critical evaluation of knowledge claims unnecessary. Finally, the researchers advocating developmental models stated that some individuals, with experience and effort, move into an evaluativist stage where knowledge is seen to have both subjective and objective aspects, highlighting the need for critical thinking skills to evaluate evidence and choose from among multiple knowledge claims.

These models are most often investigated with qualitative methods, despite the fact that the various stages, positions, or levels are often characterized in ways that appear amenable to factor-based quantitative analyses. For example, in Table 1 the "reality" descriptor that is used as one means of characterizing Kuhn et al.'s (2000) various levels could be considered a latent factor, with strong belief in a directly knowable reality on one end of the continuum and weak belief or disbelief on the other end. Likewise, the dimension for characterizing knowledge seems to range from a belief in certain and external to uncertain and internal knowledge. Hofer and Pintrich (1997) have previously outlined this overlap between developmental and systems of beliefs models.

The two main systems of beliefs models are those of Schommer-Aikins (2004) and Hofer and Pintrich (1997). These models characterize personal epistemology using more or less independent continua and measure participants' degree of belief on each continuum with quantitative methods. Whereas these authors recognize disagreement about whether some of the posited belief continua are epistemological, there seems to be general consensus that personal epistemology can be conceptualized as including beliefs about the nature of knowledge and the nature of knowing (Hofer, 2006; Muis et al., 2006; Schommer-Aikins, 2004). Named after their naïve poles, latent factors capture beliefs regarding the nature of knowledge, that is, its simplicity and certainty. The nature of knowing continua capture students' beliefs regarding the source and justification of knowledge (see Hofer & Pintrich, 1997, for a more thorough description of these continua). Studies have shown relations between individual continua and various academic outcomes (e.g., Hofer, 2000; Schommer, 1990, 1993; Schommer, Crouse, & Rhodes, 1992). Only recently have systems-of-beliefs researchers begun to suggest that the continua may develop in a coordinated, organized manner (Bendixen & Rule, 2004; Hofer, 2001, 2004a; Muis et al., 2006). Whereas no published research posits how these independent beliefs might systematically develop, a progression like that in developmental models such as Kuhn et al.'s (2000) seems reasonable.

Table 1
Kuhn et al.'s (2000) Model of Epistemological Thinking

Level	Reality	Knowledge	Critical thinking	Assertions
Realist	Directly knowable	Certain, from external source	Unnecessary	Copies of an external reality
Absolutist	Directly knowable	Certain, from external source	Used to determine who is "correct"	Facts that can be correct or incorrect
Multiplist	Not directly knowable	Uncertain, from human minds	Irrelevant	Opinions that cannot be questioned
Evaluativist	Not directly knowable	Uncertain, from human minds	Used to make good decisions and facilitate understanding	Judgments that can be evaluated

The Role of Justification in Models of Personal Epistemology

Only a few personal epistemology models include justification as a factor (Hofer & Pintrich, 1997; King & Kitchener, 2004), which is surprising given that the central question of philosophical epistemology is how individuals can reliably justify knowledge qua knowledge (Plantinga, 1993; Pollock & Cruz, 1999; Williams, 2001). When included in personal epistemology models, justification is defined with only one dimension, and those definitions tend to be little more than a recapitulation of Perry's model: "As individuals learn to evaluate evidence and to substantiate and justify their beliefs, they move through a continuum of dualistic beliefs to the multiplistic acceptance of opinions to reasoned justification for beliefs" (Hofer & Pintrich, 1997, p. 120). Philosophical epistemology identifies numerous means of justification, making it unlikely that the various ways individuals justify knowledge could be captured with a single factor or continuum (Greene et al., 2008; Murphy, 2003a, 2003b, 2005; Murphy, Alexander, Greene, & Edwards, 2005, 2007). If, like philosophers, students consider many different kinds of warrants (e.g., appeals to authorities, logic, coherence) when justifying knowledge claims, attempts to quantitatively measure those beliefs about knowing on a single continuum are unlikely to succeed. Restricting justification to a single factor may be one reason that numerous studies of personal epistemology measures have found little evidence of construct validity and poor data-model fit (Clarebout et al., 2001; DeBacker et al., 2008; Wood & Kardash, 2002).

Domain Generality Versus Specificity

There is also an argument about whether personal epistemology is domain general or domain specific (see Alexander, 2006; Hofer, 2006; Muis et al., 2006, for more complete reviews). Some researchers have argued for multiple levels of personal epistemology (see Figure 1). These range from domain specificity, to specificity at the disciplinary level, to differences between well- versus ill-structured domains only (Donald, 1990; Frederiksen, 1984) to a domain-general level (Muis et al., 2006). Empirical evidence tends to support domain specificity (Buehl & Alexander, 2005; Buehl, Alexander, & Murphy, 2002; Hallett, Chandler, & Krettenauer, 2002). As Hofer (2006) has pointed out, however, empirical investigations into the domain generality or specificity of personal epistemology must be viewed skeptically given the acknowledged problems with score reliability and validity from quantitative mea-

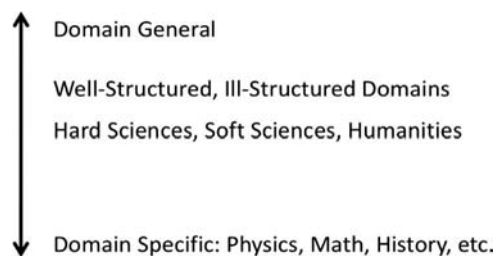


Figure 1. Range of differences regarding the domain generality or specificity of epistemic cognition.

asures of the phenomena (Clarebout et al., 2001; DeBacker et al., 2008; Wood & Kardash, 2002).

Measuring Personal Epistemology

Scores from quantitative measures of personal epistemology have often failed to meet common reliability and validity criteria. Numerous studies have utilized summed factor scores with internal reliability coefficients below the common .7 standard (DeVellis, 2003) as predictor variables, increasing the likelihood of Type I and Type II errors, as well as potential inflation and deflation of the associated regression coefficients (Pedhazur, 1997). Likewise, there have been problems with the factor structure and stability of measures of personal epistemology (Clarebout et al., 2001; DeBacker et al., 2008), suggesting problems with construct validity and item design. Such issues may lead to difficulties in demonstrating hypothesized relations among personal epistemology and other outcomes of interest, such as academic performance (Wood & Kardash, 2002). Muis et al. (2006) have argued for the use of confirmatory factor analyses (CFAs; Kline, 2005) when evaluating the construct validity of scores from measures of personal epistemology. CFAs test the hypothesized item-factor relations while also disattenuating error from the indicator items, producing better measures of the underlying constructs (Kline, 2005; Pett, Lackey, & Sullivan, 2003).

The homogeneous samples often used in studies of personal epistemology may be another cause of some of these reliability and validity problems (Greene et al., 2008). Whereas some models of personal epistemology suggest little development prior to late high school or college age (King & Kitchener, 2004; Perry, 1970), others claim that relevant variance exists among students as young as middle school age (Chandler, Hallett, & Sokol, 2002; Greene et al., 2008; Kuhn & Weinstock, 2002). Studies that use homogeneous samples of college students may be restricting the range of responses, leading to poor score reliability and limiting variance that could be used to predict outcomes such as academic performance. In general, empirical, quantitative work in personal epistemology has suffered from low predictive validity, often failing to find hypothesized relations between factors and outcomes or producing contradictory results from study to study (Schraw & Olafson, 2008).

Summary of Previous Models of Personal Epistemology and Their Measurement

An evaluation of models of personal epistemology suggests the need for further research. First, despite Hofer's (2001) call, the connections between the positions in developmental models and the more or less independent factors in systems of beliefs models have not been thoroughly investigated. Although developmental models make reference to characteristics that could be considered factors, the positions have been the focus. An integrative model has the potential to provide evidence supporting the positions utilized in qualitative research on developmental models and to explain why quantitative measures have shown contradictory results from study to study (Clarebout et al., 2001). Second, given the work in philosophical epistemology, both developmental and systems of beliefs models of personal epistemology have inadequately addressed issues of justification (see also Murphy et al.,

2007). Finally, although researchers have made strong arguments against the domain generality of personal epistemology (Alexander, 2006; Buehl et al., 2001, 2002, 2005; Hofer, 2006; Muis et al., 2006) debate continues regarding the extent of domain specificity (Schommer & Walker, 1995; Schommer-Aikins, Duell, & Barker, 2003). Previously, we presented a conceptual model of EOC that addresses each of these concerns (see Greene et al., 2008). In this study, we present empirical evidence regarding a measure derived from the conceptual model and test hypothesized relations between academic performance, educational level, and the positions in this model.

A Model of Epistemic and Ontological Cognition

Terminology

Our model of EOC represents an integration of aspects of both developmental and systems of beliefs models of personal epistemology, in answer to Hofer's (2001) call. We agree with Kitchener (2002) that the term *epistemic cognition*, which means thinking about knowledge, is preferable to terms including a version of the word *epistemology*, which literally means "the study of knowledge." Like Hofer and Pintrich (1997), we believe that beliefs about both the nature of knowledge and the nature of knowing are relevant phenomena. We argue, however, that only the latter should be called epistemic cognition. In accordance with Murphy and colleagues (Murphy, 2005; Murphy & Alexander, 2004; Murphy et al., 2007), we claim that beliefs regarding the nature of knowledge should be termed ontological cognition, not epistemic cognition.

Philosophical ontology involves the search for a complete set of mutually exclusive, necessary, and sufficient classifications, or categories, of reality (Chisholm, 1996). In the sense that we are using the term, every academic domain has a compendium of knowledge, and that knowledge is composed of a number of categories or classifications. Within a domain like science, ontologies can include classifications for objects (e.g., chromosomes, plants), processes (e.g., the freezing of water), and relations (e.g., causal, independent). Attributes typify the things within a classification. For example, all objects may have the attributes "observable" and "possessing mass." In sum, the ontology of an academic domain consists of various classifications or kinds of knowledge, and attributes differentiate these classifications. When personal epistemology researchers talk about individuals having a "simple" view of knowledge in an academic domain, they are referring to a simple knowledge ontology with a limited number of classifications. For example, a simple ontology for knowledge in history might be that all knowledge claims are "facts" and that there are no other classifications or categories. An individual with a more sophisticated ontology would recognize that there are multiple classifications (e.g., historical facts, speculations, motives), each with different attributes (Chi, 1992).

When researchers discuss the "certain" nature of knowledge, it seems that this refers to an attribute ascribed to classifications (i.e., it is "static over time") that may or may not be accurate. For example, some mathematical relations may be relatively static over time (e.g., $2 + 2 = 4$) but cosmological relations may change (e.g., views regarding geocentric versus heliocentric universes). With development and educational experience, the attributes students ascribe to classifications should become more accurate. Also, it is important to note that Chi (1997) distinguished between charac-

teristic attributes, those that are salient but often nonessential to differentiating categories (e.g., the color of an animal's fur), and core attributes, those that are often less salient but more diagnostic of category membership (e.g., being warm-blooded). A sophisticated ontology would have to be based on core, not characteristic, attributes. Overall, the development of more sophisticated beliefs regarding the nature of knowledge involves the elaboration of knowledge ontologies, that is, classifications and their attributes. Therefore, like Murphy et al. (2005, 2007), we see the nature of knowledge factors in personal epistemology as more ontological than epistemic and refer to them as such.

This is not just a matter of semantics. Viewing these phenomena as ontological also more readily allows connections to other areas of research. For example, students often make inappropriate ontological classifications that lead to confusion and resistance to conceptual change (Chi, 1992; Slotta & Chi, 2006; Vosniadou & Brewer, 1994). Students with a simple ontology in science, for example, might classify all claims as "facts" with the attribute "does not change over time." With further educational experience and cognitive development, these students come to understand that their ontology is too limited and add new classifications with different attributes. Eventually, students might experience what Slotta and Chi (2006) referred to as an ontological shift. This occurs when a knowledge claim moves from one kind of classification to another, as when science students realize that heat is a "process," not a "substance." Students who believe that knowledge in a domain is simple and certain have a limited ontology that makes inappropriate ontological classifications likely, leading to conceptual confusion. Thus, recognizing the ontological aspects of personal epistemology suggests a potentially fruitful connection with ontological training and conceptual change research.

Description of Model

Having explained terminology, we provide a brief description of the model to demonstrate our integration of positional and dimensional models. Our epistemic and ontological cognitive development (EOCD) model is composed of four positions that are similar to those used by Kuhn et al. (2000). However, the model includes an additional position advanced by Chandler et al. (2002; Hallett et al., 2002) and uses their terminology rather than Kuhn's (2005; see Table 2). Each of the four positions is described in terms of the profile of individuals' beliefs along three dimensions: Simple and Certain Knowledge, Justification by Authority, and Personal Justification.

Our Simple and Certain Knowledge dimension combines Hofer and Pintrich's (1997) and Schommer-Aikins's (2004) Simple Knowledge and Certain Knowledge factors for two reasons. First, factor analyses of Hofer and Pintrich's and Schommer-Aikins's measures suggest a single dimension, rather than separate Simple and Certain Knowledge factors (Hofer, 2000; Qian & Alvermann, 1995). Second, theoretically we believe it unlikely that someone could have a simple view of knowledge but also see it as uncertain. We think it equally unlikely that someone could have a complex view of knowledge but believe it to be static. At base, however, the dimensionality of this factor is an empirical question that has yet to be adequately addressed.

Our justification dimensions measure the degree to which an individual feels that particular sources are sufficient to warrant a knowledge claim. Individuals who have strong belief in justifica-

Table 2
Model of Epistemic and Ontological Cognition

Educational level	Ill-structured domains				Well-structured domains			
	Position	Belief in . . .			Position	Belief in . . .		
		SC	JA	PJ		SC	JA	PJ
Early elementary school	Realism	Strong	Strong	Strong	Realism	Strong	Strong	Strong
Late elementary school to early college	Dogmatism or skepticism	Weak	Strong	Weak	Realism	Strong	Strong	Strong
Middle to late college	Rationalism	Weak	Weak	Strong	Dogmatism or skepticism	Weak	Strong	Weak
Post-undergraduate education	Rationalism	Weak	Moderate	Moderate	Rationalism	Weak	Weak	Strong
						Weak	Moderate	Moderate

Note. SC = Simple and Certain Knowledge dimension; JA = Justification by Authority dimension; PJ = Personal Justification dimension.

tion by authority would say that they “know” something if an expert, teacher, or other reputable source said it. Those with a strong belief in personal justification warrant knowledge claims based on personal experience or logic.

The terms *strong agreement*, *moderate agreement*, and *weak agreement* are used to characterize individuals’ beliefs for both theoretical and practical reasons. First, we posit that differences in these beliefs are quantitative, not qualitative, in nature. Therefore, what differentiates individuals across the positions in the EOCD model is the extent of their belief that knowledge is simple and certain, not whether they hold fundamentally different beliefs. Second, the use of these terms directly corresponds to responses to Likert-type survey items measuring these beliefs. Consider a survey with a Likert-type item such as “The things we call knowledge in history will always be knowledge, they will never change,” with a response scale ranging from 1 (*completely disagree*) to 6 (*completely agree*). Given our EOCD model, we expected some individuals to respond toward the *strongly agree* end of the continuum and individuals in other positions to respond toward the *strongly disagree* end of the continuum. The former responses indicate strong agreement with the belief that knowledge is simple, whereas the latter responses can be considered weak agreement. Next, we describe how the profile of responses across the three dimensions (e.g., weak belief in Simple and Certain Knowledge, strong belief in Justification by Authority, weak belief in Personal Justification) can be used to characterize individuals in terms of their EOCD model position.

The positions in our model, and the order in which individuals progress through those positions, are similar to other work in personal epistemology (e.g., Chandler et al., 2002; Kuhn et al., 2000). The first position in the model, realism, is similar to Kuhn et al.’s absolutism, and its profile includes strong belief in Simple and Certain Knowledge, Justification by Authority, and Personal Justification. Therefore, individuals in this position see knowledge as objectively knowable, making almost any means of justification sufficient to warrant a knowledge claim. With the realization that knowledge is not simple and certain, the evaluation of warrants used to establish justification becomes necessary, and individuals move into either dogmatism or skepticism. Dogmatists rely on authority figures to provide the justification necessary to warrant their knowledge claims. Such individuals differ from Kuhn et al.’s absolutists in that they recognize that knowledge is not objectively knowable.

Skeptics, on the other hand, differ from absolutists both in their views regarding ontology and in their epistemic cognition. In this

sense, skeptics are very much like Kuhn et al.’s (2000) multiplists in believing that their own personal experience constitutes sufficient justification. Because this experience is inherently subjective, it is not comparable across individuals. Skeptics would strongly endorse the idea that justification comes from internal, idiosyncratic sources. Finally, the EOCD model is similar to other models in positing that with experience and reflection, some individuals develop from either dogmatism or skepticism into rationalists, who resemble Kuhn et al.’s evaluativists. Rationalists continue to believe that knowledge is complex and dynamic but take more nuanced positions toward means of justification. In essence, a rationalist would say, “Authority figures can sometimes justify knowledge, depending upon their credibility and whether their claims cohere with other lines of evidence.” Likewise, rationalists would moderately agree with statements claiming that personal experience or reasoning is sufficient justification, again depending on whether that experience aligns with other forms of evidence.

Therefore, each position within the model is characterized by the strength of individuals’ beliefs along both epistemic and ontological dimensions. The model also includes a level of domain specificity, grouping academic areas into well- and ill-structured domains (Donald, 1990; Frederiksen, 1984). This level of domain specificity addresses the “context” called for by Hofer. We believe that movement away from a realist position begins first in ill-structured domains such as history or literature, with movement in well-structured domains starting later. In Table 2, we provide a listing of the positions and the approximate educational level in which they are most likely to be found, acknowledging that these are educated guesses based on past work in personal epistemology and cognitive development (see Greene et al., 2008, for a more complete description of the conceptual model as well as its theoretical and philosophical foundations).

Assessing Our Model of Epistemic and Ontological Cognition

One test of a conceptual model’s adequacy is whether it can lead to an instrument with strong psychometric qualities. Thus, we developed a quantitative instrument to measure the postulated relations among dimensions and positions in this model. The reliability and validity of scores (i.e., the hypothesized relations among indicators and the factors they were designed to capture) from such an instrument were examined with a CFA as were our

hypotheses regarding domain specificity. However, the conceptual model also includes hypotheses regarding how individuals' developmental positions are constituted by their profiles of scores across the latent factors. A statistical test of these relations between positions and latent factor scores requires the use of factor mixture modeling (Gagné, 2006; Lubke & Muthén, 2005).

Factor mixture analysis is analogous to a combination of CFA and latent profile analysis (LPA, also called latent class cluster analysis; see Magidson & Vermunt, 2004). LPA is like an analysis of variance (ANOVA) in which group membership is unknown and must be determined from the data. Qualitatively different groups (or clusters) of individuals are determined a posteriori from their profile of scores across a set of continuous indicators, such as responses to Likert-type items on a survey. Factor mixture modeling differs from LPA in that in the former, the indicators are the latent factor scores from the CFA, rather than the observed item responses, which are used in the latter. Use of latent factor scores allows for the disattenuation of the error from the individual item responses, resulting in stronger indicators of the latent classes.

In a factor mixture model, multiple classes are posited, each with its own measurement model. Usually, restrictions are imposed regarding the invariance of various parameters across the classes' measurement models. These restrictions are similar to the assumption of homogeneity of variance in an ANOVA analysis. In most factor mixture models, the factor loadings and intercepts are fixed to be equal across classes, resulting in what Meredith (1993) called strong factor invariance. In some factor mixture models, an additional constraint is imposed beyond strong invariance: fixed residual variances and covariances across classes, referred to as *strict* factor invariance. In addition, factor variances and covariances can be left free to vary across classes or constrained to be equal. Testing factor mixture models involves fitting multiple models with various levels of constraints (e.g., strict and strong invariance, fixed and free to vary factor variances and covariances) as well as different numbers of latent classes. The relative fit of various factor mixture models positing different numbers of latent classes and various forms of invariance can be compared using information criteria (e.g., Akaike's information criterion [AIC], Bayesian information criterion [BIC]; Vermunt & Magidson, 2002) as well as various forms of the likelihood ratio test (Lo, Mendell, & Rubin, 2001; Nylund, Asparouhov, & Muthén, 2007). It is important to note, however, that factor mixture modeling requires a measurement model with strong psychometric qualities (Lubke & Muthén, 2005), both to have faith in profile interpretations and to allow a full test of the conceptual model.

Factor mixture modeling is particularly helpful in testing our EOCD model because it can indicate whether individuals' patterns of scores across the latent factors (i.e., their latent profile) match those hypothesized in the model (see Table 2). For example, if the underlying conceptual model is correct, the factor mixture model analysis should reveal a cluster of realists whose latent factor means (i.e., the average of all of the latent factor scores among participants in the cluster) are all relatively high on the Simple and Certain Knowledge, Justification by Authority, and Personal Justification factors. On the other hand, a cluster of dogmatists would have a different profile: Their latent factor means would be relatively low on the Simple and Certain Knowledge and Personal Justification factors but relatively high on the Justification by Authority factor. The factor mixture model can also produce

evidence disconfirming parts of our conceptual model. If a profile of means was found that had not been hypothesized in Table 2 (e.g., high on Simple and Certain Knowledge but low on both epistemic cognition factors), this would be evidence that the conceptual model needed revision. Factor mixture modeling also allows for testing developmental hypotheses about EOC by regressing the latent profiles onto educational level. Finally, the factor mixture model can be used to test hypotheses about the relations between EOCD model positions and academic performance.

Thus, the EOCD model, which is informed by philosophy and posits specific relations among latent factors and positions, can be used as the basis for a quantitative instrument. Then, scores from this instrument can be statistically tested with a factor mixture model to determine whether the results support the conceptual model. This quantitative approach to capturing the relations among positions and factors provides some of the same strengths as more qualitative techniques, where participant responses across various dimensions are used to construct profiles (see Table 1).

Whether the positions in the EOCD model have predictive validity is an empirical question. The first step toward assessing the model's predictive validity is to examine the relations among positions and self-report measures of educational level and course grades. Finding hypothesized relations among academic performance, educational level, and position is a step toward confirming the EOCD model.

Limitations of the Assessment Techniques

As with any new model and measure, more research will be required to build the case for the validity of the EOCD model. For example, we claim that skeptics are unlikely to accept knowledge claims warranted by experts, preferring to use their own more personal means of evaluating knowledge. Additional support for this claim, beyond self-report data, could be obtained by identifying skeptics using the Epistemic and Ontological Cognition Questionnaire (EOCQ) and then testing whether they actually behave in a skeptical manner in a discovery learning environment by recording what sources of justification they privilege (e.g., the results of in-class experiments) and which ones they ignore (e.g., expert analyses in textbooks).

Further, Muis et al. (2006) have suggested that the most advanced kinds of epistemic cognition may be difficult to characterize on Likert-type response scales with dualistic and multiplistic positions as their endpoints. In addition, we (Greene et al., 2008) have addressed the possibility that individuals are unable to accurately report their beliefs about the nature of knowledge and knowing. Overall, whether personal epistemology can be measured using questionnaires and Likert-type items is an empirical question that has yet to be addressed. If psychometric analyses of quantitative instruments continue to produce poor results, this may be evidence against questionnaire methods in general, necessitating the use of other methods, such as qualitative interviews.

The Current Study

One of the primary goals of this study was to test a quantitative instrument derived from the EOCD model to determine whether it had strong psychometric qualities and yielded scores supporting the

integration of positional and dimensional aspects of EOC. In addition, we sought to show “proof of concept” for modeling EOC as an integration of dimensions and positions. This means showing that taking latent factor mean profiles into account results in better data-model fit and predictive utility than relying on factor scores alone.

In terms of the specific positions and dimensions in the EOC model, this study’s findings would support our model if the latent classes inferred from the data had factor score profiles matching those posited in Table 2. For example, we expected to find at least one latent class whose members were characterized as having relatively strong belief in Simple and Certain Knowledge, Justification by Authority, and Personal Justification, as compared with individuals in other latent classes. This latent class would then be characterized as composed of realists. Overall, with factor mixture modeling, we expected evidence (i.e., at least one latent class with the posited factor score profile) for each of the four EOC model positions (i.e., realism, dogmatism, skepticism, rationalism).

The EOCQ was administered to students ranging from middle school through graduate school, with the hope of avoiding restriction of range problems. The EOCQ contained parallel items measuring EOC in mathematics and history to test our hypothesis of difference in EOC between well- and ill-structured domains. The first set of hypotheses concerned the adequacy of the proposed measurement model (see Figure 2), whereas the second set addressed the factor mixture model (see Figure 3).

Measurement Model

CFA was used to test hypotheses regarding the psychometric qualities of the EOCQ, including the reliability and validity of its scores, and evidence of domain specificity. The first set of hypotheses consisted of the following:

Hypothesis 1: Scores from the EOCQ will have sufficient construct reliability to warrant further analysis of the data.

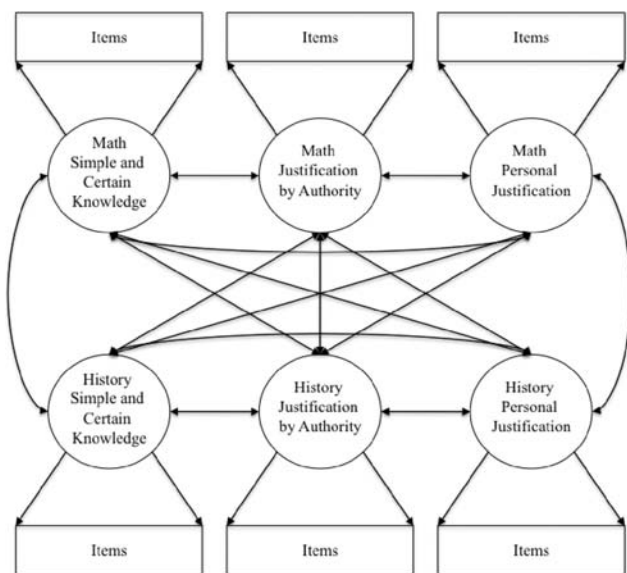


Figure 2. Hypothesized domain-specific measurement model. Residual covariances between parallel items are not shown for the sake of clarity.

Hypothesis 2: Scores from the EOCQ will demonstrate evidence of construct validity, producing CFA data-model fit indices that meet or exceed the criteria recommended by Hu and Bentler (1999).

Hypothesis 3: A domain-specific measurement model will show statistically significantly better data-model fit than measurement models with domain-general factors.

Factor Mixture Model

The second set of hypotheses concerned the relations among the latent factors, the developmental positions they indicated, and the educational level of the respondents (see Figure 3).

Hypothesis 4: The factor mixture models, allowing for latent factor mean differences across clusters, will have better data-model fit than the measurement model, as judged by the preponderance of evidence among the information criteria (Nylund et al., 2007).

Hypothesis 5: The clusters determined using the factor mixture model will have profiles of latent factor means that match those hypothesized for the four positions in the conceptual model.

Hypothesis 6: Within each latent cluster, the position for the ill-structured domain (history) will be at least as advanced as the position for the well-structured domain (mathematics).

Hypothesis 7: The covariates, educational level and mathematics and history grades, will be statistically significantly related to the latent cluster positions, with higher levels of education and better grades predictive of more sophisticated positions.

Hypotheses 4 through 7 test the adequacy of the instrument and, by extension, the conceptual model underlying it. However, Hypothesis 4 is also a kind of “proof of concept” regarding the possibility of simultaneously modeling developmental positions and systems of beliefs. Hypothesis 7 tests the utility or predictive validity of such modeling.

Method

Participants

A total of 740 participants took part in this study, of whom 68% were female. Among the sample, 127 were in middle school, 173 were in high school, 305 were undergraduates, and 135 were graduate students. The middle school and high school students were recruited from schools in a suburb of Philadelphia while the undergraduate students attended a state university in the mid-Atlantic region. There were two graduate student samples: one from the previously mentioned state university and the second from a state university of similar quality in the South. The undergraduate and graduate samples included students with more than 70 majors. Participants were solicited in their classrooms, and the response rate for consent forms and completed surveys was 65%.

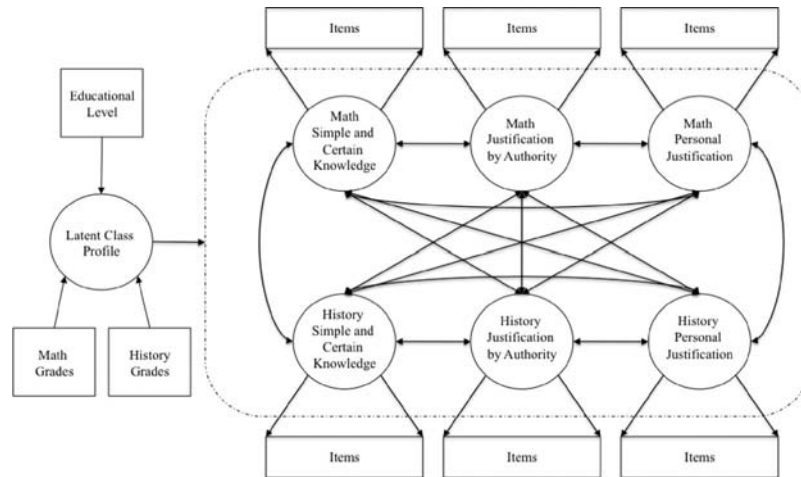


Figure 3. Factor mixture model. Residual covariances between parallel items are not shown for the sake of clarity.

Informed consent was obtained from all participants 18 years or older and from the parents or guardians of participants under the age of 18 years. None of the participants received any kind of incentive (e.g., course credit) for participating in the study.

Measure: The EOCQ

The EOCQ was refined during a pilot study conducted prior to this study (see Greene, 2007; Greene, Torney-Purta, Azevedo, & Robertson, in press). The version of the EOCQ used in this study contains two sets of 13 items each, one pertaining to mathematics and the other pertaining to history (see Appendix A for a complete list of items). The Simple and Certain Knowledge, Justification by Authority, and Personal Justification factors were captured with five, four, and four items, respectively. The EOCQ item response scale ranges from 1 to 6, as follows: 1 (*completely disagree*), 2 (*mostly disagree*), 3 (*somewhat disagree*), 4 (*somewhat agree*), 5 (*mostly agree*), and 6 (*completely agree*). Ten of the items were phrased so that higher ratings indicated stronger beliefs, whereas three of the items were phrased so that higher ratings indicated weaker beliefs. The EOCQ also includes seven demographic items, bringing the total number of items on the instrument to 33.

Procedure

Middle school and high school participants completed the EOCQ in their classes. Undergraduate and graduate students received an addressed envelope containing the EOCQ and the consent form and were asked to return the completed forms via campus mail. These students received all instructions via a letter attached to the survey. All participants were told, either verbally or in the letter, "This questionnaire is meant to assess your understanding of different academic and non-academic areas. There are no right or wrong answers. Please answer the items honestly, as your responses will be confidential." Participants filled out the EOCQ individually and were asked to not discuss the questionnaire with others while completing it. Participants were told to "do the best [he or she] can" in interpreting the items, and content

clarification questions were not answered so as to prevent bias. After collection of the EOCQ, participants were asked to refrain from discussing the questionnaire with any peers who might be taking it at another time.

Data Analysis

The data analysis techniques, CFA and factor mixture modeling, were described previously. Two remaining issues included treatment of missing data and the quantification of the covariates, educational level and mathematics and history grades.

Missing data. Only 32 of the 740 completed EOCQ surveys contained missing data. There were 21 unique missing data patterns, but no pattern was repeated more than four times. Given the low percentage of total data missing and the variability in missing data patterns, these data were treated as missing completely at random (Little & Rubin, 2002), and full information likelihood estimation (FIML) was used. There are many benefits to utilizing FIML rather than casewise deletion, listwise deletion, or mean imputation, including more accurate and efficient estimates of parameters than other methods and the advantage of being able to include cases with partial missing data in the analysis (see Enders, 2001).

Educational level. Participants were coded as in middle school, high school, college, or graduate school. College students were further asked whether they were a freshman, sophomore, junior, or senior. For analysis, the educational level was operationalized as the number of years in school and treated as a continuous variable. Therefore, middle school was coded as 6, high school as 11, university freshman as 13, sophomore as 14, junior as 15, senior as 16, and graduate school as 17 (i.e., 17 years of schooling, kindergarten through 12th grade plus four years of college and at least one year of graduate school). We focused on educational level rather than age as a predictor of EOC because much of the literature has supported exposure to education, and not biological age, as a major catalyst for development (Bendixen & Rule, 2004; Hofer & Pintrich, 1997).

Mathematics and history grades. We also asked students to self-report the grades they “usually get” in mathematics classes and also in history classes. The response set for each item was the same, with seven options: Almost all As, mostly As with some Bs, Almost all Bs, Mostly Bs with some Cs, almost all Cs, mostly Cs with some Ds, and Almost all Ds or Fs. We coded these responses as 1 through 7, respectively, and treated these variables as continuous in the analysis. The mean for mathematics grades was 2.37, with a standard deviation of 1.28, and the mean for history grades was 2.20, with a standard deviation of 1.11. We decided against gathering grades from academic records, as this would require identifying students by name, complicating the institutional review board process and potentially lowering the response rate.

Results

The first set of results refers to the measurement model and Hypotheses 1 through 3. The second set of results involves the factor mixture model and addresses the remaining hypotheses. All analyses were performed using Mplus 5.0 (Muthén & Muthén, 2008).

Measurement Model Results

Hypothesis 1: Construct reliability. The first hypothesis was that scores on the dimensions would display acceptable levels of reliability, and there was evidence to support this claim. We determined construct reliability using Coefficient *H* (Hancock & Mueller, 2001). Three of the construct reliabilities were over .7; two were just below this figure; and one, for mathematics Simple and Certain Knowledge, was well below .7 (see Table 3). Using DeVellis’s (2003) guidelines, these findings suggest acceptable reliability of scores for all but the last factor.

Hypothesis 2: Construct validity. The second hypothesis was that the relations in the data would support the conceptual model, the EOCQ model. Across a number of metrics, there was evidence that the EOCQ model (see Figure 2) had acceptable fit to the data. Using CFA, we obtained a measurement model standardized root mean square residual (SRMR) of .08 and a robust root mean square estimate of approximation (RMSEA) of .06, meeting the criteria for acceptable model–data fit recommended by Hu and Bentler (1999). The robust chi-square for the model was statistically significant, $\chi^2(271, N =$

740) = 1,029.368, $p < .001$. All but two of the unstandardized factor loadings were statistically significant at the .01 level. Items written to load negatively on their proposed factors did so. The absolute value of 19 of the 26 standardized factor loadings was .4 or higher (see Table 4). Factor correlations and variance extracted information can be found in Table 3.

Overall, the results for Hypotheses 1 and 2 suggested that the scores for all of the factors except mathematics Simple and Certain Knowledge had acceptable reliability and construct validity. The problems with the mathematics Simple and Certain Knowledge items, in particular, the low standardized factor loadings, pointed to the need for further “bootstrapping” (King & Kitchener, 2004, p. 11) between item validation, item design, and the EOCQ model. Item revision is warranted prior to future administrations of the EOCQ.

Hypothesis 3: Domain specificity. The third hypothesis dealt with the claim that EOC was domain specific, meaning that it would vary depending on the academic area being considered. Overall, there was evidence supporting this hypothesis. Three domain-general measurement models were run to test the hypothesis of domain specificity of factors. The first domain-general measurement model was the same as the model used in Hypothesis 2 (see Figure 2) except that it had all of the mathematics and history Simple and Certain Knowledge items loading on a single factor rather than two domain-specific factors. This domain-general measurement model was nested in the EOCQ model; thus, the two could be compared using a scaled chi-square difference test (Satorra & Bentler, 1999). This test was statistically significant, scaled $\chi^2_{\text{diff}}(10) = 245.56$, $p < .001$, indicating support for the superiority of the data–model fit of the domain-specific measurement model. Similar models were run for the Justification by Authority and Personal Justification factors, and the subsequent scaled chi-square difference tests were both statistically significant, scaled $\chi^2_{\text{diff}}(9) = 342.10$, $p < .001$, scaled $\chi^2_{\text{diff}}(9) = 622.64$, $p < .001$.

Factor Mixture Model Results

The second set of hypotheses concerned the relations among the latent factors, the developmental positions they indicated, and the covariates: educational level, mathematics grades, and history grades (see Figure 3).

Hypothesis 4: Factor mixture model fit. In this study, we wished to show “proof of concept” that modeling a profile of

Table 3
Measurement Model Correlations, Construct Reliability (*H*), and Variance Extracted

Variable	Mathematics			History		
	Simple and Certain Knowledge	Justification by Authority	Personal Justification	Simple and Certain Knowledge	Justification by Authority	Personal Justification
Mathematics factor						
Simple and Certain Knowledge	.451	.189	–.379**	.015	.027	.043
Justification by Authority		.819	–.175**	.210**	.404**	–.080*
Personal Justification			.771	.076	.045	–.082
History factor						
Simple and Certain Knowledge				.697	.635**	–.712**
Justification by Authority					.899	–.415**
Personal Justification						.678
Variance extracted	.127	.521	.368	.279	.677	.308

Note. Construct reliability estimates (*H*) are presented on the diagonal.
* $p < .05$. ** $p < .01$.

Table 4
Measurement Model Standardized Factor Loadings

Mathematics factor			History factor		
Simple and Certain Knowledge	Justification by Authority	Personal Justification	Simple and Certain Knowledge	Justification by Authority	Personal Justification
-.533**	.672**	.730**	-.489**	.758**	.643**
.124	.681**	.287**	.357**	.811**	.019
.428**	.782**	.807**	.705**	.857**	.610**
.331**	.747**	-.454**	.648**	.862**	-.669**
-.138**			-.334**		

** $p < .01$.

participants' scores across the three dimensions was superior to examining their scores on the dimensions independently. Whereas a full discussion of the technical aspects of fitting the factor mixture models (see Figure 3) can be found in Appendix B, overall the results do show proof of concept: Modeling EOC in an integrated manner, with dimensions used to identify positions, had a better fit to the data than modeling the dimensions independently (see Table 5). Next are details regarding the construct validity and reliability of the factor mixture model determined to be the best fit to the data, referred to as the *final* factor mixture model.

All of the unstandardized factor loadings across classes in the final factor mixture model were statistically significant, except for one item, measuring the history Personal Justification factor, which was also statistically nonsignificant in the measurement model. The standardized factor loadings ranged in absolute value from .032 to .988, with 108 of the 156 loadings (70%) above .4 (see Table 6). Factor correlations, construct reliabilities, and variance extracted by class are shown in Table 7. The absolute values of the factor correlations ranged from .05 to .72, but none was so high as to suggest a lack of discriminant validity between factors.

Coefficient H values for 19 of the 36 (53%) factors were above .7, and the variance extracted ranged from .086 to .811, with 16 of the 36 (44%) above .4. Overall, looking at both the reliability and validity of factor scores across classes, the mathematics and history Simple and Certain Knowledge factors were most problematic, whereas the mathematics and history Justification by Authority factors had the strongest psychometric qualities. Again, these findings are informative regarding how well the items on the EOCQ performed across the latent classes and can be used to "bootstrap" future item and conceptual model refinement.

Hypothesis 5: Evidence for conceptual model. The next hypothesis, for which there was mixed support, concerned the interpretation of the latent factor means within each class to determine whether the posited relations between profiles and positions were supported (see Table 2). Each class had a latent factor mean for each dimension (Simple and Certain Knowledge, Justification by Authority, Personal Justification) based on the average of latent factor scores for all participants in that class (see Table 8). There are numerous ways these latent class factor means can be compared with one another and subsequently used to characterize

Table 5
Factor Mixture Model Fit Indices

Number of classes/Invariance	Log-likelihood ^a	AIC ^b	BIC ^b	SABIC ^b
1 (measurement model)	-28,810.718	57,833.436	58,321.741	57,985.153
2				
Strict	-28,160.310	56,552.620	57,084.456	56,716.122
Strong	-27,704.568	55,729.137	56,439.780	55,947.609
3				
Strict	-28,025.734	56,303.467	56,881.151	56,481.064
Strong	-27,484.869	55,377.738	56,313.035	55,665.275
4				
Strict	-27,995.463	56,262.924	56,886.455	56,454.615
Strong	-27,310.214	55,126.428	56,286.380	55,483.031
5				
Strict	-27,952.522	56,197.044	56,866.424	56,402.831
Strong	-27,220.984	55,045.968	56,430.575	55,471.636
6 ^c				
Strict	-27,919.239	56,150.479	56,865.706	56,370.360
Strong	-27,126.735	54,955.470	56,564.731	55,450.203

Note. AIC = Akaike's information criterion; BIC = Bayesian information criterion; SABIC = sample-size-adjusted BIC. Italic numbers indicate the model with the best fit according to that fit index.

^a Models with log-likelihood values closer to zero are considered a better fit to the data. ^b Models with lower information criteria values are considered a better fit to the data. ^c This model exhibited numerous estimation problems, including Heywood cases and latent factor correlations greater than one.

Table 6
Factor Mixture Model Standardized Factor Loadings

Factor	Mathematics factor			History factor		
	Simple and Certain Knowledge	Justification by Authority	Personal Justification	Simple and Certain Knowledge	Justification by Authority	Personal Justification
Mathematics						
Simple and Certain Knowledge	-.221 ^{**} , -.288 ^{**} , -.237 ^{**} , -.279 ^{**} , -.264 ^{**} , -.245 ^{**}	.565 ^{**} , .790 ^{**} , .504 ^{**} , .728 ^{**} , .751 ^{**} , .703 ^{**}	.455 ^{**} , .820 ^{**} , .769 ^{**} , .788 ^{**} , .640 ^{**} , .608 ^{**}	-.284 ^{**} , -.531 ^{**} , -.478 ^{**} , -.497 ^{**} , -.836 ^{**} , -.278 ^{**}	.621 ^{**} , .870 ^{**} , .624 ^{**} , .620 ^{**} , .724 ^{**} , .770 ^{**}	.461 ^{**} , .754 ^{**} , .472 ^{**} , .639 ^{**} , .778 ^{**} , .478 ^{**}
Justification by Authority	.183 ^{**} , .260 ^{**} , .182 ^{**} , .214 ^{**} , .255 ^{**} , .257 ^{**}	.512 ^{**} , .773 ^{**} , .560 ^{**} , .843 ^{**} , .717 ^{**} , .803 ^{**}	.207 ^{**} , .422 ^{**} , .300 ^{**} , .278 ^{**} , .285 ^{**} , .243 ^{**}	.199 ^{**} , .274 ^{**} , .216 ^{**} , .191 ^{**} , .233 ^{**} , .288 ^{**}	.686 ^{**} , .898 ^{**} , .699 ^{**} , .779 ^{**} , .794 ^{**} , .799 ^{**}	.032, .062, .041, .037, .041, .047
Personal Justification	.427 ^{**} , .697 ^{**} , .455 ^{**} , .966 ^{**} , .501 ^{**} , .758 ^{**}	.580 ^{**} , .856 ^{**} , .626 ^{**} , .934 ^{**} , .759 ^{**} , .832 ^{**}	.560 ^{**} , .903 ^{**} , .852 ^{**} , .938 ^{**} , .635 ^{**} , .711 ^{**}	.448 ^{**} , .669 ^{**} , .552 ^{**} , .472 ^{**} , .812 ^{**} , .623 ^{**}	.722 ^{**} , .923 ^{**} , .756 ^{**} , .715 ^{**} , .784 ^{**} , .906 ^{**}	.414 ^{**} , .740 ^{**} , .465 ^{**} , .644 ^{**} , .626 ^{**} , .440 ^{**}
History						
Simple and Certain Knowledge	.394 ^{**} , .585 ^{**} , .427 ^{**} , .504 ^{**} , .592 ^{**} , .464 ^{**}	.590 ^{**} , .942 ^{**} , .527 ^{**} , .758 ^{**} , .824 ^{**} , .878 ^{**}	-.317 ^{**} , -.481 ^{**} , -.369 ^{**} , -.451 ^{**} , -.299 ^{**} , -.466 ^{**}	.346 ^{**} , .524 ^{**} , .542 ^{**} , .464 ^{**} , .988 ^{**} , .410 ^{**}	.760 ^{**} , .911 ^{**} , .724 ^{**} , .772 ^{**} , .714 ^{**} , .928 ^{**}	-.373 ^{**} , -.703 ^{**} , -.459 ^{**} , -.848 ^{**} , -.837 ^{**} , -.438 ^{**}
Justification by Authority	-.092 [*] , -.163 [*] , -.110 [*] , -.114 [*] , -.114 [*] , -.116 [*]			-.242 ^{**} , -.433 [*] , -.277 ^{**} , -.243 ^{**} , -.228 ^{**} , -.294 ^{**}		

Note. Multiple values within a cell indicate standardized factor loadings for that item for each latent class, in order, from Class 1 through Class 6.

* $p < .05$. ** $p < .01$.

Table 7
Factor Mixture Model Factor Correlations, Construct Reliability (H), and Variance Extracted by Class

Factor	Mathematic factor			History factor		
	Simple and Certain Knowledge	Justification by Authority	Personal Justification	Simple and Certain Knowledge	Justification by Authority	Personal Justification
Mathematics						
Simple and Certain Knowledge	.334, .623, .371, .935, .508, .639	.443 ^{**}	-.550 ^{**}	-.111	.154	.050
Justification by Authority		.849, .932, .646, .922, .853, .894	-.412 [*]	.191 [*]	.624 ^{**}	-.095
Personal Justification			.467, .875, .813, .903, .609, .661	-.100	-.201 ^{**}	.088
History						
Simple and Certain Knowledge				.366, .654, .556, .497, .978, .525	.526 ^{**}	-.723 ^{**}
Justification by Authority					.799, .947, .801, .824, .845, .933	-.352 ^{**}
Personal Justification						.390, .778, .454, .798, .819, .437
Variance extracted	.086, .201, .098, .265, .150, .186	.410, .710, .309, .672, .583, .651	.166, .474, .386, .445, .256, .288	.100, .253, .185, .156, .488, .161	.489, .811, .493, .525, .570, .728	.131, .403, .163, .386, .425, .154

Note. Coefficient H values are presented on the diagonal. When multiple values appear in a cell, they indicate the value for each class, in order, Class 1 through Class 6.
** $p < .01$.

Table 8
Latent Class Factor Means and Interpretations

Measure	Mathematics factor			History factor		
	Simple and Certain Knowledge	Justification by Authority	Personal Justification	Simple and Certain Knowledge	Justification by Authority	Personal Justification
Class 1						
Latent mean	−0.630	−0.399	0.893	−0.365	−0.125	0.293
Relative strength of belief	Weak	Strong	Strong	Strong	Strong	Weak
Profile interpretation		Unclear ^a			Unclear ^a	
Class 2						
Latent mean	−0.424	−0.408	−0.079	−1.152	−0.909	0.832
Relative strength of belief	Weak	Weak	Strong	Weak	Strong	Moderate
Profile interpretation		Skeptic			Unclear ^a	
Class 3						
Latent mean	−0.556	−0.833	−0.651	−1.719	−1.267	0.824
Relative strength of belief	Weak	Weak	Strong	Weak	Moderate	Moderate
Profile interpretation		Skeptic			Rationalist	
Class 4						
Latent mean	0.109	0.098	−0.512	−1.816	−1.225	1.586
Relative strength of belief	Strong	Strong	Strong	Weak	Weak	Strong
Profile interpretation		Realist			Skeptic	
Class 5						
Latent mean	−1.337	−0.712	0.334	−2.851	−1.802	1.802
Relative strength of belief	Weak	Weak	Strong	Weak	Weak	Strong
Profile interpretation		Skeptic			Skeptic	
Class 6 ^b						
Latent mean	0.00	0.00	0.00	0.00	0.00	0.00
Relative strength of belief	Strong	Strong	Strong	Strong	Strong	Weak
Profile interpretation		Realist			Unclear ^a	

^a These profiles of factor means were not posited in the conceptual model.

^b The means in one class must be set to zero to establish reference points for the other means, as they are all measured on a latent scale (Kline, 2005).

the latent classes. Once characterized, these latent classes can be compared with the positions posited in the EOCD model as another means of testing the adequacy of this conceptual model of EOC.

One way to characterize the latent classes is to examine a plot of the latent class means across the six dimensions (see Figure 4). This figure gives a general sense of how the latent class factor means vary across each dimension, as well as the pattern, or profile, of means across a class. Clearly, the classes' latent factor means differ within each factor, and the classes differ in their latent factor mean profiles. Another method of characterizing the latent classes is to perform a series of independent *t* tests between each pair of latent class factor means within each dimension. Latent class factor means that statistically significantly differ from one another give some indication of the relative position of each class on the latent dimension continuum and can be used to identify whether the class participants had strong, moderate, or weak belief in that dimension (see Table 2). The results of these independent samples *t* tests, accounting for unequal sample sizes, can be found in Appendix C.

Both a subjective analysis of the pattern of latent class factor means (see Figure 4) and the results of the statistical tests revealed an interpretable set of profiles. Each latent class factor mean could be categorized as indicating weak, moderate, or strong agreement with its dimension. For example, the latent class factor means within the history Simple and Certain Knowledge dimension (see Table 8) consisted of two (Latent Classes 1 and 6) that could be characterized as strong agreement that knowledge in history was

simple and certain. The other four latent class factor means were all statistically significantly lower than those for Latent Classes 1 and 6 (but did not statistically significantly differ from each other; see Appendix C), which was interpreted as weak agreement with the belief that knowledge in history was simple and certain. Similar techniques were used to characterize the rest of the latent class factor means for the other dimensions, mostly relying on the results of the statistical tests but also relying on our judgment when those tests did not point to a clear characterization of weak, moderate, or strong agreement.

Once differences across latent classes were determined within dimensions, the pattern of beliefs across dimensions within latent classes could be examined. These patterns of latent class factor means, also called profiles, can then be compared with those posited in the EOCD model and shown in Table 2. For example, within the mathematics factors for Class 2, the pattern of weak, weak, and strong, across the Simple and Certain Knowledge, Justification by Authority, and Personal Justification factors, respectively, matched the skeptic position posited in the conceptual model. In all, eight of the 12 profiles matched those posited in the conceptual model and can be considered evidence of the validity of that conceptual model. The four profiles not posited by the model, as indicated in Table 8, are not supportive of the EOCD model and suggest that further bootstrapping between model revision and item design is necessary. We address possible explanations of these findings later in this article. Overall, the fact that a majority of the latent class profiles were interpretable provided further evidence of the adequacy and utility of the final factor mixture

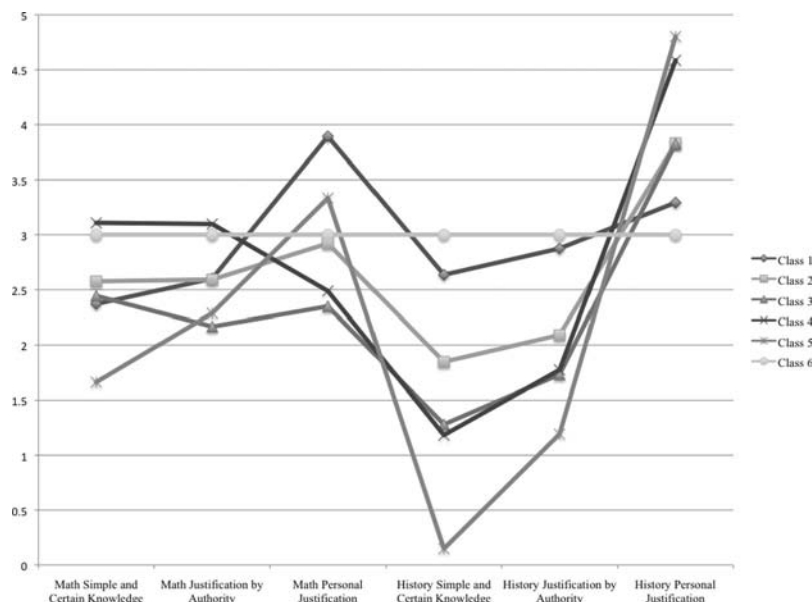


Figure 4. Latent class factor means. All latent class factor means in this figure were adjusted by adding three to their value, recentering the scale within the positive number range.

model (Lubke & Muthén, 2005). In addition, within five of the six classes, the presence of different positions for mathematics and history provide further evidence of the domain specificity of EOC.

Hypothesis 6: Relation between well- and ill-structured domains. It was hypothesized that for each latent class, the history position inferred from the data would be at least as advanced (i.e., more sophisticated in terms of the EOCD model; see Table 2) as that of the mathematics position, and overall this hypothesis was supported. In Classes 3, 4, and 5, the history position inferred from the latent class factor means is at least as advanced, if not more so, than the mathematics position. The interpretation of the history position in Class 6 is unclear. In Class 6, the mathematics position is realist; therefore, whatever the history position is, it is at least as sophisticated as the realist position, because nothing is less sophisticated than realist in the EOCD model. The positions for math and history in Class 1 are difficult to interpret, so this class provides no evidence regarding Hypothesis 6. Class 2 is also problematic, but the history position includes weak agreement with Simple and Certain Knowledge. Weak agreement with Simple and Certain Knowledge is indicative of a dogmatist, skeptic, or rationalist position, all at least as sophisticated as the mathematics skeptic position. Therefore, the results of these analyses provide evidence supporting Hypothesis 6.

Hypothesis 7: Relations with covariates. We hypothesized that the positions in our EOCD model would be related to educational level and academic performance. In general, our findings showed that participants with more educational experience and higher grades in mathematics and history were more likely to be identified as having more advanced EOC. As estimated as a part of the final factor mixture model, the latent class variable is regressed on the covariates, educational level, mathematics grades, and history grades. Because the latent class variable is categorical, this is a multivariate multinomial logistic regression (Hosmer & Leme-

show, 2000; Lubke & Muthén, 2005), and the results are shown in Appendix D.

The pattern of results for educational level showed that undergraduate and graduate students were statistically significantly more likely to be in Classes 2 through 5 and less likely to be in Classes 1 or 6. Although Classes 1 and 6 were more problematic to interpret, in general, these classes could be considered less “sophisticated” because for the most part they had latent factor means that indicated strong agreement with the Simple and Certain Knowledge factor. According to the conceptual model, these scores indicate more of a realist position. Latent Classes 2 through 5 were generally characterized as having more sophisticated positions in both mathematics and history.

Higher values on this mathematics grades variable indicated poorer performance, and the results showed that for every one unit increase on this covariate, participants were statistically significantly more likely to be in Class 1 as opposed to Classes 2, 4, 5, or 6. This also aligns with the conceptual model if Class 1 is interpreted as indicating less sophisticated beliefs. Finally, the history grade covariate was scored in the same way as the mathematics grade covariate, and results showed that for every one unit increase on this variable, participants were statistically significantly more likely to be in Class 6 than Class 3, again aligning with the conceptual model if Class 6 is interpreted as a less sophisticated class than Class 3.

Discussion

We have attempted to answer Hofer’s (2001) call for an integration of developmental and systems of beliefs models of personal epistemology that could also account for possible domain specificity in students’ cognition. Our integrated conceptual model, the EOCD model, incorporates ideas from philosophy and outlines one ontolog-

ical and two epistemic dimensions that capture students' beliefs about the nature of knowledge and knowing. The EOCD model also describes specific relations between the profile of students' beliefs across these three dimensions and their position within the model, allowing for positional differences in EOC across domains. Using this model, in this study we sought to measure EOC in a quantitative manner, developing an instrument with strong psychometric qualities including scores that were reliable and showed evidence of construct and predictive validity. A complete test of the integrated conceptual model required factor mixture modeling, which could simultaneously model dimensional and positional aspects of EOC as well as allow for a test of the psychometric qualities of our instrument and our hypotheses of domain specificity between mathematics and history. This simultaneous modeling resulted in a positional classification related to three covariates of interest: educational level and mathematics and history grades. Finally, we sought to use this analysis as the first step toward bootstrapping between item design, validation, and conceptual model revision.

In general, our findings support both our conceptual model and the utility of simultaneously modeling both the dimensional and positional aspects of epistemic and ontological cognition. The measurement model had good data-model fit by Hu and Bentler's (1999) standards, with acceptable levels of construct reliability for all but one of the factors. The data produced evidence supportive of a domain-specific model of epistemic and ontological cognition, in line with other empirical findings and theoretical arguments (Alexander, 2006; Buehl et al., 2002, 2005; Hofer, 2006; Muis et al., 2006). Although the CFA, dimensions-only model did have acceptable data-model fit, a six-latent-class factor mixture model, which used the dimensions as indicators of positions, provided a better fit across numerous statistical metrics and subjective standards. We interpreted this finding as proof of concept for the benefits of simultaneously modeling dimensional and positional aspects of students' EOC. Likewise, these findings provide support for our EOCD model, in particular, for specific hypotheses regarding how the three epistemic and ontological factors "progress in reasonably predictable directions, activated in context, operating as epistemic cognition" (Hofer, 2001, p. 377). Eight of the twelve positions in the factor mixture model were posited in the conceptual model, and in each latent class, the history position was at least as sophisticated as the mathematics position, as predicted. Finally, the covariates, educational level, mathematics grades, and history grades, were all statistically significantly related to participants' latent class, demonstrating evidence for the predictive validity of the model.

We were encouraged that eight of the 12 latent classes had profiles of scores on the three belief dimensions that matched those posited in the EOCD model. The two latent classes of realists had relatively strong belief in Simple and Certain Knowledge, Justification by Authority, and Personal Justification (see Table 8). These findings not only align with the EOCD model but also with other developmental models of personal epistemology, including King and Kitchener's (2004) prereflective thinkers and Kuhn and Weinstock's (2002) realists. In this study, realists were also more likely to have less educational experience and lower grades in mathematics and history than other participants, supporting the idea that realist beliefs are not as beneficial in learning environments as those of skeptics, dogmatists, and rationalists.

Participants in five of the latent classes displayed a profile indicative of skeptics, including relatively weak belief in Simple

and Certain Knowledge and Justification by Authority, and relatively strong belief in Personal Justification. Again, these findings align with the EOCD model as well as with King and Kitchener's (2004) quasireflective thinkers and Kuhn and Weinstock's (2002) multiplists. The skeptic profile was the most frequent in the sample, which makes sense given that the majority of participants were in adolescence or early adulthood, a time during which beliefs about knowledge and knowing become relativistic, according to many researchers.

There was one latent class identified as rationalist, which is not surprising given that more advanced levels of epistemic and ontological cognition tend to be rare overall (King & Kitchener, 1994; Perry, 1970). These rationalists were more likely to have a great deal of educational experience and tended to have higher grades in history than did the other participants. Although the data are cross-sectional, therefore limiting our ability to make developmental claims, for five of the six latent classes, EOC varied by domain. This suggests that individuals do experience a kind of Piagetian horizontal decalage (Hofer & Pintrich, 1997) in the development of their EOC. Domain-general models and measures that ignore this variability are most likely underestimating participants' EOC, as Chandler et al. (2002) have suggested.

These findings indicate that the EOCD model aligns with other conceptual models of personal epistemology and shed light on important concerns in the personal epistemology literature, including debates regarding construct definition, the domain generality or specificity of those constructs, how best to model those constructs (i.e., positional or dimensional), and concerns regarding the psychometric adequacy of quantitative measures of students' beliefs about knowledge and knowing. However, there are limitations to this study, including the sample size, characteristics, and response rate; issues concerning the functioning of items in the EOCQ; ambiguity of some of the latent class positions inferred from the analyses; and the possibility that EOC may include dimensions that were not examined in this study.

Limitations

Sample size, characteristics, and response rate. Factor mixture models often require a very large sample size to converge (Lubke & Muthén, 2005). This study involved 740 participants, beyond the norm in research on personal epistemology, but not large in terms of work with factor mixture models. Although the models did converge, they did so only when factor variances and covariances were constrained across classes (see Appendix B). This could point to a need for a larger sample.

The sample consisted of participants who were currently enrolled in formal education settings. It may be that students who elect to leave the formal education system, especially those not attending college, engage in different types of epistemic and ontological cognition that are not well-captured in this sample. Further, both the middle school and high school were located in upper-middle-class areas. Finally, whereas we are not aware of any systematic bias in terms of those who did not elect to respond to our study, a more diverse and representative sample would bolster claims of external validity.

Item functioning. Items were revised on the basis of the findings from the pilot study of the EOCQ (Greene, 2007; Greene et al., in press). However, judging from the standardized factor

loadings for the measurement and factor mixture models, two of the simple and certain knowledge items and one of the personal justification items needed to be revised.

Given that two of the five Simple and Certain Knowledge items failed to meet common psychometric standards, it is not surprising that the factors associated with these items, both in mathematics and history, had the lowest construct reliability, and variance extracted values of the six factors. These factor scores may be substandard indicators of students' ontological cognition, impeding the ability to discriminate between realists and dogmatists, for example. Poor item functioning may be due to the ways in which individual items are written or may be indicative of the difficulty of measuring personal epistemology using questionnaire methods (Hofer, 2004a; Muis et al., 2006).

Latent class positions. Whereas a majority of the profiles of latent class factor means matched those posited in the conceptual model, four did not. It may be that the item functioning problems described previously caused these aberrant profiles. If this were the case, item revision, but not conceptual model revision, would be required. However, it may be that the conceptual model itself needs to be amended and that some or all of these empirically derived profiles need to be incorporated into the model as accurate descriptions of students' EOC. This second option seems most likely for the history profiles in latent Classes 1 and 6 (strong, strong, weak; see Table 8). That the same unexpected profile was found across two classes warrants further investigation. This profile is similar to the hypothesized dogmatist position (weak, strong, weak; see Table 2) in terms of the justification factors, which had stronger psychometric qualities than the ontological factor. It may be that dogmatists maintain a simple and certain ontology of knowledge in academic domains and differ from realists only in that they reject personal means of justification. It is also important to note that the hypothesized dogmatist position was not revealed in any of the six latent classes, further strengthening the idea that the strong, strong, weak profile is indeed indicative of a true dogmatist position. If this is the case, then the conceptual model should be amended. Interviews with students may shed light on these issues.

Finally, statistical tests of the models in this study provide support for the idea that the dimensions develop "in reasonably predictable directions" (Hofer, 2001, p. 337). However, whereas the analyses suggest that relations among the dimensions are more powerful predictors of the covariates than the individual dimensions, it may be the case that these dimensions are in fact relatively independent, as Schommer-Aikins (2004) suggested. The presence of four unclear latent classes, and the absence of a dogmatist latent class, may be signs that analyses of the influence of each individual dimension on outcomes of interest should be performed rather than positional analyses. This issue can be addressed by comparing the predictive utility of treating the dimensions as a profile versus using them as individual predictors of outcomes of interest like academic performance.

Dimensions of EOC. Finally, we have made the case that EOC is multidimensional, and in particular we have argued that more than one type of justification must be measured. However, the dimensions we have chosen to study may not completely capture the phenomena of epistemic and ontological cognition. A full understanding of an individual's cognition may require more than the measurement of Justification by Authority and Personal Justification (see Murphy et al., 2007, for other types of justification). Likewise, some of the items used to capture our Simple and Certain knowledge factors did not

perform well. This may indicate that measuring ontological cognition requires several additional dimensions. For example, in terms of historical ontologies, individuals may vary in the degree to which they see speculation about the intentions of historical figures as knowledge *qua* knowledge. Those who restrict the term *historical knowledge* to facts and events may have a somewhat more limited ontology than those who use assumptions about intentionality as a way of explaining historical actions. Measuring these differences may allow for a more reliable and valid differentiation between realists and other positions.

Future directions. Several items on the EOCQ require revision. Cognitive interviewing (Karabenick et al., 2007) should be used to vet new items and ensure that they accurately capture the abstract and complex nature of ontological cognition. Likewise, interviews could inform changes to the EOCQ model itself, including investigation of dogmatists' ontology of academic domains.

In general, ontological cognition is difficult to operationalize, as it involves multiple categories and attributes that may differ across academic areas (Greene et al., 2008; Murphy et al., 2007; Slotta & Chi, 2006). For example, there may be categories in history (e.g., empathetic inferences; VanSledright, 2004) that simply do not exist in mathematics. Such differences may make it difficult to create a measure with parallel items across such domains. Modeling common positions using differing dimensions and items may be a way to capture ontological cognition across domains. Further, it may be the case that additional justification dimensions are needed to fully capture the ways individuals think about knowing. The relevance of these dimensions may differ by domain as well, making parallel measures even less likely to have strong psychometric qualities.

Further, more work is needed regarding the degree of domain specificity of EOC (see Figure 1). The current findings argue against a domain-general approach, but our contention that domain differences exist only between well- and ill-structured domains, and not at a more specific level, remains to be tested. Future work should include parallel measures of multiple well- and ill-structured domains.

Through this conceptual model, we argue that development in EOC can begin as early as late elementary school. However, our understanding of the process of development is still provisional (Hofer, 2008). Educational level is not much more than a proxy for other processes. Bendixen and Rule (2004) have suggested that those processes involve epistemic doubt, volition, and conceptual change, and these promising ideas should be investigated. Further longitudinal work is needed to assess what processes influence development. It is clear, however, that those studies need to begin with students much younger than college age.

Finally, these findings suggest that analyses that focus on relations between positions and dimensions, rather than the individual dimensions alone, will result in a better fit to the data as well as stronger and more consistent predictive utility. This suggests that research in classrooms (e.g., Conley, Pintrich, Vekiri, & Harrison, 2004; Schommer, 1990) and intervention studies may continue to produce confusing or contradictory results if they focus on relating academic outcomes to individual dimensions. The results of this study, along with Hofer's call for integration, converge to make a strong argument for depending on aggregates or profiles of scores across dimensions as the predictors of academic outcomes. We hope that such studies will indicate which outcomes are influenced by EOC, as well as which

student, classroom, and contextual factors influence the development of students' beliefs about the nature of knowledge and knowing.

Conclusion

Personal epistemology is a relatively new and complex area of research, requiring multiple investigations utilizing numerous techniques to explore the problem space. We advocate for both quantitative and qualitative investigations into students' beliefs about knowledge and knowing and hope that the results of such work can be integrated into a coherent and useful conceptual model. We believe our EOCD model represents a step forward in that it models both dimensional and positional aspects of the phenomena; makes specific, testable claims regarding their development over time; and aligns with research in educational psychology as well as work from philosophy. This study provides initial support for the conceptual model and identifies avenues to explore.

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Appendix A

List of Items From the Epistemic and Ontological Cognition Questionnaire

Simple and Certain Knowledge Items

In [math/history],^a the truth means different things to different people.^b

To know [math/history] well, you need to memorize what you are taught.

In [math/history], what is a fact today will be a fact tomorrow. [Mathematicians'/Historians'] knowledge of the facts about math does not change.

[Math/History] is so complex that humans will never really understand it.^b

Justification by Authority Items

If a [mathematician/historian] says something is a fact, I believe it.

Things written in [math/history] textbooks are true.

I believe everything I learn in [math/history] class.

If a [math/history] teacher says something is a fact, I believe it.

Personal Justification Items

In [math/history], everyone's knowledge can be different because there is no one absolutely right answer.

In [math/history], if you believe something is a fact, no one can prove to you that you are wrong.

In [math/history], what's a fact depends upon a person's point of view.

[Mathematical/Historical] knowledge is all factual and there are no opinions.^b

^a Each item was presented twice in its entirety, once with the math term and once with the history term.

^b These items were worded in a manner opposite the other items.

Appendix B

Elaboration of the Methods Used to Fit and Test the Factor Mixture Models

A series of factor mixture models were estimated. Lubke and Muthén (2007) have found that the inclusion of covariates can facilitate model estimation; therefore, the latent class variable was regressed on the three covariates for each model (see Figure 3). Individual models were run with and without the factor variances and covariances constrained to be equal across classes. Likewise, separate models imposing strong and strict invariance (Meredith, 1993) were run. In addition, factor mixture models with two through six classes were tested. Models positing seven classes did not terminate normally, suggesting that attempting to model more than six latent classes was not warranted with these data (Lubke & Muthén, 2005). Therefore, the seven class models are not reported. In total, 20 different models were fit to the data, crossing factor variance and covariance restrictions (2: constrained or no) with type of invariance (2: strong or strict) with number of proposed classes (2 through 6). Models allowing the factor variances and covariances to vary across classes exhibited numerous problems, including Heywood cases (Kline, 2005) and latent factor correlations greater than one. Estimation problems such as these suggest that these models do not adequately fit the data (Lubke & Muthén, 2005); therefore, they are also not reported here. Similar problems occurred when attempting to estimate a six-class, strict invariance model. All of the other models (see Table 5 for a list of models identified by number of classes and type of invariance) converged to admissible solutions using user-generated start values, but not

the randomly perturbed start values generated by Mplus 5.0. It appeared that these models were very sensitive to start values.

Models can be compared using the log-likelihood, and the information criteria fit indices Akaike's information criterion, Bayesian information criterion (BIC), and the sample-size adjusted BIC. Better fitting models have a log-likelihood closer to zero, and for each of the information criteria, the model with the lowest value, including negatives, has the best fit (Vermunt & Magidson, 2002). Lo-Mendell-Rubin likelihood ratio test (LMR; Lo et al., 2001) and the bootstrap likelihood ratio test (bLRT; Nylund et al., 2007) values of less than .05 indicated support for a model with at least as many classes as the one tested. For model comparison purposes, a measurement model is equivalent to a one-class factor mixture model. Taking into account the log-likelihood and information criteria metrics of data-model fit, the six-class factor mixture model with strong factorial invariance was best, with only the BIC indicating otherwise (see Table 5). The LMR and bLRT were then computed for this six-class model, with the LMR not supporting a six-class model, but the bLRT supporting it. The LMR has been criticized as being less accurate than the bLRT (Nylund et al., 2007); therefore, we relied more on the latter test. Thus, across all of the various criteria for data-model fit, we felt that the six-class strong variance factor mixture model provided the best fit to the data and used it as the final factor mixture model for all further analyses.

(Appendices continue)

The factor mixture model also produces probabilities regarding the likelihood of each participant being in each of the six classes. These probabilities can be combined into an entropy value (Magidson & Vermunt, 2004). The entropy value is an indication of how well class membership is predicted by the indicators and thus is another means of assessing the adequacy and utility of the model. This value varies from zero to one, with higher values indicating better prediction. The entropy value for the final factor mixture model was .823. Assigning participants to the class for which they had the highest probability resulted in latent classes with 123, 127, 139, 93, 46, and 196 members, respectively.

As expected, the standardized factor loadings in the final factor mixture model differed from those in the measurement

model. This can be attributed to the fact that when there exist mean differences in factor scores across unidentified classes, as can be the case in a one-class measurement model, the relations between items and factors can be attenuated. Therefore, the factor mixture model, which models these mean-level differences, can result in changes among these relations between the measurement and factor mixture models. In addition, the standardized factor loadings also varied across classes in the final factor mixture model. This is due to the fact that whereas strong factorial invariance constrains unstandardized factor loadings to be equal across classes, the residual variances were free to vary across classes, resulting in standardized factor loadings that differed across classes.

Appendix C

Independent Sample *t* Tests of Latent Class Factor Means by Dimension

Latent class factor mean comparison	Latent factor mean difference	<i>t</i> test value
Mathematics: Simple and Certain Knowledge		
Class 1 versus 2	−0.206	−1.250
Class 1 versus 3	−0.074	−0.267
Class 1 versus 4	−0.739	−8.031**
Class 1 versus 5	0.707	2.774**
Class 1 versus 6	−0.630	−3.198**
Class 2 versus 3	0.132	0.380
Class 2 versus 4	−0.533	−2.565*
Class 2 versus 5	0.913	1.873
Class 2 versus 6	−0.424	−1.208
Class 3 versus 4	−0.665	−1.864
Class 3 versus 5	0.781	0.997
Class 3 versus 6	−0.556	−1.201
Class 4 versus 5	1.446	4.679*
Class 4 versus 6	0.109	0.470
Class 5 versus 6	−1.337	−2.096*
Mathematics: Justification by Authority		
Class 1 versus 2	0.009	0.013
Class 1 versus 3	0.434	0.720
Class 1 versus 4	−0.497	−0.553
Class 1 versus 5	0.313	0.175
Class 1 versus 6	−0.399	−0.494
Class 2 versus 3	0.425	5.312**
Class 2 versus 4	−0.506	−4.345**
Class 2 versus 5	0.304	1.481
Class 2 versus 6	−0.408	−1.672
Class 3 versus 4	−0.931	−13.884**
Class 3 versus 5	−0.121	−1.123
Class 3 versus 6	−0.833	−5.444**
Class 4 versus 5	0.810	5.314**
Class 4 versus 6	0.098	0.447
Class 5 versus 6	−0.712	−2.125*
Mathematics: Personal Justification		
Class 1 versus 2	0.972	1.331
Class 1 versus 3	1.544	1.543
Class 1 versus 4	1.405	1.371
Class 1 versus 5	0.559	0.357
Class 1 versus 6	0.893	1.215
Class 2 versus 3	0.572	0.830
Class 2 versus 4	0.433	0.755
Class 2 versus 5	−0.413	−0.619

Appendix C (*continued*)

Latent class factor mean comparison	Latent factor mean difference	<i>t</i> test value
Class 2 versus 6	-0.079	-0.176
Class 3 versus 4	-0.139	-0.144
Class 3 versus 5	-0.985	-0.655
Class 3 versus 6	-0.651	-0.957
Class 4 versus 5	-0.846	-0.861
Class 4 versus 6	-0.512	-0.800
Class 5 versus 6	0.334	0.601
History: Simple and Certain Knowledge		
Class 1 versus 2	0.787	4.714**
Class 1 versus 3	1.354	4.268**
Class 1 versus 4	1.451	7.164**
Class 1 versus 5	2.486	8.682**
Class 1 versus 6	-0.365	-1.229
Class 2 versus 3	0.567	1.866
Class 2 versus 4	0.664	3.477**
Class 2 versus 5	1.699	6.367**
Class 2 versus 6	-1.152	-4.100**
Class 3 versus 4	0.097	0.247
Class 3 versus 5	1.132	1.651
Class 3 versus 6	-1.719	-3.737**
Class 4 versus 5	1.035	3.494**
Class 4 versus 6	-1.816	-5.389**
Class 5 versus 6	-2.851	-7.790**
History: Justification by Authority		
Class 1 versus 2	0.784	0.515
Class 1 versus 3	1.142	0.654
Class 1 versus 4	1.100	0.550
Class 1 versus 5	1.677	0.478
Class 1 versus 6	-0.125	-0.110
Class 2 versus 3	0.358	0.433
Class 2 versus 4	0.316	0.489
Class 2 versus 5	0.893	1.151
Class 2 versus 6	-0.909	-1.765
Class 3 versus 4	-0.042	-0.040
Class 3 versus 5	0.535	0.323
Class 3 versus 6	-1.267	-1.733
Class 4 versus 5	0.577	0.684
Class 4 versus 6	-1.225	-1.966*
Class 5 versus 6	-1.802	-5.347**
History: Personal Justification		
Class 1 versus 2	-0.539	-3.418**
Class 1 versus 3	-0.531	-3.627**
Class 1 versus 4	-1.293	-8.248**
Class 1 versus 5	-1.509	-5.003**
Class 1 versus 6	0.293	0.942
History: Personal Justification		
Class 2 versus 3	0.008	0.069
Class 2 versus 4	-0.754	-6.698**
Class 2 versus 5	-0.970	-4.564**
Class 2 versus 6	0.832	3.341**
Class 3 versus 4	-0.762	-7.192**
Class 3 versus 5	-0.978	-4.880**
Class 3 versus 6	0.824	3.552**
Class 4 versus 5	-0.216	-1.625

(Appendices continue)

Appendix C (continued)

Latent class factor mean comparison	Latent factor mean difference	<i>t</i> test value
Class 4 versus 6	1.586	8.113**
Class 5 versus 6	1.802	5.347**

* $p < .05$. ** $p < .01$.

Appendix D

Multinomial Logistic Regression of Latent Class Variable on Covariates

Class comparison ^a / Independent variable	Estimate (<i>b</i>)	<i>e^b</i> (change in odds) ^a
Class 1 versus 2		
Educational level	-0.702**	0.496**
Mathematics grades	0.571**	1.770**
History grades	0.387	1.473
Class 1 versus 3		
Educational level	-0.515*	0.598*
Mathematics grades	0.297	1.346
History grades	0.580	1.786
Class 1 versus 4		
Educational level	-0.602*	0.548*
Mathematics grades	0.740**	2.096**
History grades	0.537	1.711
Class 1 versus 5		
Educational level	-1.345**	0.261**
Mathematics grades	0.441*	1.554*
History grades	0.357	1.429
Class 1 versus 6		
Educational level	-0.268	0.765
Mathematics grades	0.381**	1.463**
History grades	0.118	1.126
Class 2 versus 3		
Educational level	0.187	1.206
Mathematics grades	-0.274	0.760
History grades	0.193	1.213
Class 2 versus 4		
Educational level	0.100	1.105
Mathematics grades	0.170	1.185
History grades	0.151	1.163
Class 2 versus 5		
Educational level	-0.643*	0.526*
Mathematics grades	-0.130	0.878
History grades	-0.029	0.971
Class 2 versus 6		
Educational level	0.434*	1.544*
Mathematics grades	-0.190	0.827
History grades	-0.268	0.765
Class 3 versus 4		
Educational level	-0.087	0.917
Mathematics grades	0.443	1.557
History grades	-0.042	0.959
Class 3 versus 5		
Educational level	-0.830**	0.436**
Mathematics grades	0.144	1.155
History grades	-0.223	0.800
Class 3 versus 6		
Educational level	0.247	1.281
Mathematics grades	0.084	1.087
History grades	-0.461**	0.630**

Appendix D (continued)

Class comparison ^a / Independent variable	Estimate (<i>b</i>)	<i>e</i> ^b (change in odds) ^a
Class 4 versus 5		
Educational level	−0.743**	0.476**
Mathematics grades	−0.299	0.742
History grades	−0.180	0.835
Class 4 versus 6		
Educational level	0.335	1.398
Mathematics grades	−0.360	0.698
History grades	−0.419	0.658
Class 5 versus 6		
Educational level	1.077**	2.937**
Mathematics grades	−0.060	0.941
History grades	−0.239	0.788

^a Results indicate the change in odds of being in the first-listed class versus the second-listed class.

^{*}*p* < .05. ^{**}*p* < .01.

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